H&M customer retention

Interview prep: every question I could be asked, with answers I can say out loud.

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Python, SQL (DuckDB), scikit-learn,
Tableau Public, Ollama. Free tools only.

How to use this

Do not memorise word for word. Learn the shape of each answer, say it out loud, and use your own words.

1. Learn the story

Say the 60 second version until it feels like talking, not reciting.

2. Know the numbers

Fourteen numbers cover almost every follow up. Learn them cold.

3. Practise the questions

Read a question, answer out loud, then compare with the page.

The basics

Data and traps

Retention and loyalty

Segments

Revenue at risk

The churn model

Dashboard and automation

Recommendations and the A/B test

Tough questions

Things never to say

Questions to ask them

Phrases to keep ready

The story

The 60 second version, then the 2 minute version for when they say tell me more.

60 seconds

H&M is losing ground to Zara and Shein, so I asked a simple question: which customers are drifting away, and how much money is at risk? I took the public H&M dataset from Kaggle, about 31.8 million purchases, and worked on a 10% sample of customers so everything runs on my laptop. I built a retention funnel, compared groups like age and newsletter, sorted every customer into on track, slipping, drifting or gone quiet, and tested that sorting on past data. Then I trained a churn model. The surprise was that the model that predicts best is not the one that protects the most money, so I ranked customers by risk times what they spend. I finished with a Tableau dashboard, a script where a free local AI writes a plain English summary, three recommendations and an A/B test design. I also kept a decision log, because I made mistakes along the way and I wanted to show how I caught them.

2 minutes

Start with the question, because it keeps the project honest. H&M is a household name that is losing ground to faster, cheaper rivals. The useful version of that for a retention team is: who is slipping, what is it worth, and what should we do first?

The data is 1.37 million customers, 105,542 products and 31.8 million purchases from September 2018 to September 2020. I sampled 10% of customers, 137,571 buyers with 3.2 million purchases, and kept each person's full history so their buying pattern stays intact. I checked the sample against the full data and the club share and average price matched.

In the funnel, 42.3% of new customers came back within 180 days and only 7.2% became loyal. Across all buyers, the loyal group, 5 or more shopping days, is 39.5% of buyers but 84.1% of revenue. So losing a loyal customer hurts far more than losing a one time buyer.

I then gave every established customer a status by comparing their silence with their own usual gap between visits. I tested it on three past dates: customers marked on track bought again in the next 90 days about 63 to 67% of the time, customers marked gone quiet only 31 to 33%. Up to 21.5% of yearly revenue sits with drifting or gone quiet customers, and about 12.1% is likely gone for good.

The churn model reached an AUC of 0.77 on a later, unseen period, against 0.66 for my simple rule. But when I looked at money, the model's riskiest customers were small spenders. Ranking by risk times past spend catches 49.4% of the revenue that later goes quiet, using the same top 20% list. That became recommendation number one, with a randomised A/B test to check whether contacting them actually works.

Everything runs from scripts, and a local AI model (Ollama) writes the monthly summary. I check every number it writes against a fact sheet, because my first version mixed up customers and revenue.

Numbers to know cold

If you can say these without looking, most follow up questions answer themselves. Revenue is an index, never pounds.

`1.37 million / 31.8 million`	Customers and purchases in the full data (Sept 2018 to Sept 2020)
`137,571 / 3.2 million`	Buyers and purchases in my 10% sample
`39.5% and 84.1%`	Loyal buyers (5+ shopping days): share of buyers, share of revenue
`42.3% and 7.2%`	New customers who came back within 180 days, and who became loyal
`39 days`	Median time to a second visit (three quarters of returners come back within 93 days)
`32.2% vs 48.6%`	Repeat rate: truly new customers vs customers back after a long break
`63 to 67% vs 31 to 33%`	Bought again in 90 days: on track vs gone quiet (three past dates)
`21.5% and 12.1%`	Revenue at risk (upper limit) and likely gone for good
`27.4%`	Peak share of revenue at risk, April 2020
`0.769 vs 0.656`	Model AUC (logistic) vs my simple rule
`49.4% vs 11.0%`	Revenue caught by risk times spend vs the plain model, same top 20% list
`49.5% vs 40.8%`	New customers back in 180 days: regular newsletter readers vs none (a link, not proof)
`26.1% vs 43.6%`	PRE-CREATE vs ACTIVE club members who came back in 180 days
`About 4,300`	People per group needed in the A/B test to see a 3 point rise in buyers

The basics

Warm up questions. Keep these short and confident.

Why did you build this project?

I am an MSc Marketing student moving into analytics, and I wanted one project that covers the whole chain: messy data, SQL, cohorts, segments, a model, a dashboard, and a recommendation someone could act on. Five small notebooks would not show that. This one starts with a business question and ends with a decision.

What was the business question?

Which H&M customers are drifting away, how much revenue is at risk, and do the loyalty club and the fashion newsletter really keep people buying? The context is that H&M is losing ground to Zara and Shein.

Why H&M data?

It is real, large and messy, with customers, products and purchases, and everyone knows the brand, so the story needs no explaining. The honest limits are that it comes from a 2022 Kaggle competition, it ends in September 2020, and the prices are scrambled, so my revenue figures are an index and never pounds.

Why only a 10% sample? Isn't that cutting corners?

I sampled customers, not rows, using a fixed rule on the customer ID, so every sampled person keeps their full history. Retention depends on gaps between visits, so cutting random rows would break exactly what I am measuring. Then I checked the sample against the full data: the average price was 0.0279 in both, and the club share matched.

It also made everything fast enough to run on a laptop, which is how an analyst would start anyway. The next step would be to run the same scripts on the full data.

Which tools did you use, and why?

Python with pandas for cleaning, DuckDB to run SQL directly on the files, scikit-learn for the model, Tableau Public for the dashboard, and Ollama for a free local AI summary. The rule I set was free tools only.

On Power BI: I have not built in it. Power BI Desktop only runs on Windows and publishing needs a paid licence, so I used Tableau. The ideas are the same: a data source, measures, charts, a one page dashboard.

> Do not claim Power BI. Saying you chose Tableau for a reason sounds stronger than a fake claim that falls apart in one follow up.

Data and traps

This is where interviewers find out if you think or just run code.

What data quality problems did you find?

Age was missing for about 15,900 customers and 77 customers had ages above 90, up to 99. I kept ages from 16 to 90 and set the rest to blank, labelled Unknown. Club status and newsletter had missing values too, so I labelled those Unknown instead of dropping people. The newsletter column also had slightly different spellings of the same label, which I cleaned.

The biggest quirk was not an error: prices are scrambled by the competition, so I treat revenue as an index.

What is a shopping day, and why not just count orders?

The data is one row per item. If someone buys four items on one afternoon, that is one visit, not four. If I counted rows, a single big shop would make someone look loyal. A shopping day is one calendar day with at least one purchase.

What is the left-censoring trap?

The data starts in September 2018. A customer whose first purchase I see in October 2018 may have shopped at H&M for years before that, I just cannot see it. So I only call someone new if their first purchase is between 20 September 2019 and 26 March 2020. That gives at least a year of silence before them in the data, and a full 180 days of follow up before the data ends.

If you remember one trap, make it this one. It shows you check what the data cannot see.

Your early cohorts looked much worse, 44% down to 30% repeat. What happened?

I noticed the repeat rate falling across early cohorts and almost reported it as a decline. Then I checked who was in those cohorts. Customers who came back after a long break repeat at 48.6%, while truly new customers repeat at 32.2%. The early cohorts were mixed with returning customers, so the trend was an artefact. I relabelled them and rebuilt the new customer funnel from the clean start date.

This is also your answer to tell me about a mistake you caught.

What does revenue index mean?

The competition scrambled the prices, so the numbers only make sense relative to each other. I report shares, like 12.1% of yearly revenue, never pounds. Saying that out loud is part of honest framing.

Retention and loyalty

Cohorts, funnels and what loyal means.

How did you define loyal, and why 5?

Five or more shopping days. I tested other cut offs. At 3 days that is 53.9% of buyers and 91.4% of revenue. At 8 days it is 27.3% and 73.5%. At 13 days it is 15.9% and 57.1%. Five days gives 39.5% of buyers holding 84.1% of revenue: enough people in the group to trust the numbers, and enough history to call it a habit.

It is a judgement call, not a law of nature, so I showed the other cut offs next to it.

What does the funnel tell you?

Out of 18,122 new customers, 42.3% came back within 180 days, and only 7.2% became loyal in that time. That means only 17% of people who return go on to become loyal. The biggest leak is right at the start: 57.7% of new customers do not come back within 180 days.

When do people come back?

The median is 39 days. A quarter come back within 10 days and three quarters within 93 days. So most second visits happen inside about three months, which is where I would put the first retention message.

Segments

Who keeps buying, and what you can and cannot say about it.

Which groups come back more?

Among new customers, 16 to 24 year olds come back 47.1% of the time, with a margin of plus or minus 1.3 points, against 35.3% for people over 65 (plus or minus 3.4). ACTIVE club members come back 43.6%, PRE-CREATE members only 26.1%. Regular newsletter readers come back 49.5% against 40.8% for people with no newsletter. First purchase channel makes almost no difference: 43.2% and 41.8%, and their margins overlap.

Does the newsletter work?

I can say it is linked to more repeat buying. I cannot say it causes it. People who sign up for a fashion newsletter already like the brand, and club and newsletter status are snapshots from the end of the data. Inside ACTIVE members the gap still holds, 50.0% against 42.0%, so it is not just club status, but it is still selection. Only an A/B test, where people are chosen at random, can show cause.

Linked to, not caused by. Say it exactly like that. It is the line they are listening for.

What do the plus or minus numbers mean?

They are 95% margins. With 5,717 people in the 16 to 24 group the margin is 1.3 points, so the true rate is probably between 45.8% and 48.4%. Small groups get wide margins, and that is the point: it tells the reader how far to trust each bar.

Why did you hide groups under 500 people?

Because tiny groups swing wildly. Eight people in the LEFT CLUB group showed 12.5% with a margin of plus or minus 22.9 points. A chart with that bar would mislead. I flag anything under 500 as too small to trust and leave it off the dashboard.

What does PRE-CREATE mean?

I do not know for sure, and the dataset does not explain it. My guess is customers who have not fully activated their club membership, but that is an assumption, so I treated it as a label and flagged it. I would ask H&M what it means before building a campaign on it.

Saying I do not know, here is my guess, and here is how I would find out beats bluffing every time.

Revenue at risk

The part a business reader cares about most.

How do you decide someone is drifting away?

I compare how long they have been silent with their own usual gap between visits. The ratio is days since last purchase divided by their median gap, with the gap never smaller than 30 days. Under 1 is on track, 1 to 2 is slipping, 2 to 4 is drifting, 4 and above is gone quiet. I only do this for established customers with at least 3 shopping days.

Using their own gap matters. Someone who shops every two weeks and goes quiet for three months is a very different story from someone who shops twice a year.

How do you know those groups mean anything?

I backtested them at three past dates. Customers marked on track bought again in the next 90 days 63% to 67% of the time. Customers marked gone quiet did so only 31% to 33%. Slipping and drifting sat in between, in order, at every date.

Explain 21.5% against 12.1%.

21.5% of yearly revenue belongs to customers who are now drifting or gone quiet. That is the upper limit of what is exposed. But some of them come back. Looking at history, 67.9% of gone quiet customers and 50.8% of drifting customers stayed gone. Weighting by revenue gives about 12.1% likely gone for good. So: up to 21.5% exposed, about 12.1% probably lost.

> Never say we are losing 21.5%. Say exposed, up to 21.5%.

What does the trend over time show?

The share of revenue at risk was about 20% in September 2019, climbed to 27.4% in April 2020, fell back to about 20% by June, and ended at 21.5%. The timing sits in spring 2020, so the pandemic period could be part of it, but I cannot prove that from this data and I would not claim it.

The churn model

Expect the sharpest questions here. Slow down and use plain words.

What did you predict, and how did you build it?

I predicted whether a customer makes no purchase in the next 90 days. For each past cut date, the features came only from data up to that date: past shopping days, days since the last purchase, their usual gap, activity in the last 90 and 180 days, spend, items, share of channel 1 purchases, how long they have been a customer, and age. I trained on two cut dates, 24 December 2019 and 24 March 2020, and tested on a third, 24 June 2020.

Why not a random train and test split?

Because the same customer shows up at several dates and because time has an order. A random split lets the model peek at the future. Testing on a later period the model never saw is how it would work in real life.

What results did you get? What is AUC?

AUC was 0.769 for logistic regression, 0.768 for gradient boosting, and 0.656 for my simple ratio rule. AUC answers one question: if I pick one customer who left and one who stayed, how often does the model give the leaver the higher risk score? 0.5 is a coin flip and 1.0 is perfect. So 0.77 means about 77 times out of 100.

It is not accuracy, so I never say the model is 77% accurate.

This is the most common trap. Practise saying it out loud.

Both models score the same. Which did you pick?

Logistic regression. It performs the same as gradient boosting here, and it is simpler, faster, and easier to explain to a manager. When a complex model gives no gain, I take the simple one.

Why did you leave club status and newsletter out of the model?

Those are snapshots from the end of the data. Club status can change after someone leaves, so using it could leak information from the future into the prediction. I kept them for the segment analysis, where I label the result as a link, and left them out of the model on purpose.

Can you trust the percentages the model gives?

For ranking, yes. For exact percentages, a little less. In the riskiest 10% it predicted 85.1% would go quiet and 81.1% did. In the safest 10% it predicted 8.4% and 6.0% did. It slightly over predicts, but the order is right. So I use it to rank customers, not to quote exact odds.

What mattered most in the model?

Number of past shopping days by a long way: scrambling it dropped the AUC by 0.079. Days since the last purchase was second, at 0.023. I add a caveat that several columns carry overlapping information, so these numbers are a rough guide.

What is the revenue twist?

The model's riskiest 20% catches 33.9% of the people who go quiet, but only 11.0% of the revenue they would have taken, less than my simple rule at 17.4%. The reason is that the riskiest customers are small spenders: they spent 0.046 on average last year against 1.898 for the safest group, about 40 times less.

So I ranked by risk times past spend. The same 20% list now catches 49.4% of the revenue that goes quiet, though only 18.7% of the people. And I stay careful with the word caught. In that list 41.9% still went quiet, so catching revenue is not the same as saving it. That is what the A/B test is for.

This is your strongest story. A better predictor can protect less money. Lead with it.

Dashboard and automation

Tableau, the local AI, and the mistakes you caught.

Why Tableau?

It is free, runs on my Mac, and publishes a public link. Power BI Desktop only runs on Windows and sharing needs a paid licence. I know Tableau uses the same building blocks as Power BI: a data source, measures, charts and a one page layout. I would be comfortable picking up Power BI quickly.

What does the dashboard show?

Three headline numbers, the new customer funnel, share of revenue by status, who comes back by age, club, newsletter and channel with margins, the at risk trend, and a comparison of targeting methods by revenue caught. It uses six small summary tables, not customer level data.

Tell me about a mistake you caught while building it.

Two. First, my trend chart showed a spike to about 41.7 in September. The bar at the bottom said 12 marks for 13 months, which gave it away. Tableau had grouped by month name, so September 2019 (20.2) and September 2020 (21.5) were added together. I switched to month with the year, and the real peak is 27.4 in April 2020.

Second, the methods chart showed 33.9 for the model, which is people caught, instead of revenue caught at 11.0. It made the best method look the worst. I swapped the column and the 49.4% bar became the longest. Both times a number looked odd and I checked it before it went in front of anyone.

Tell me about the AI summary script.

A script refreshes the results, builds a short fact sheet, and asks a local model, llama3.2 running in Ollama, to write a plain English update. It is free and the numbers never leave my laptop. The script then checks every number in the text against the fact sheet and flags cause and opinion words.

My first version said 58.2% of customers were on track when it is revenue, called the newsletter a positive effect, and said the contact list had been used when it was only a test on past data. The number check still passed, because the numbers were real and the meaning was wrong. So I tightened the prompt, added fixed honesty notes, and I read the summary myself every time.

The lesson to say out loud: a check catches one kind of error, so a human still reads the output.

Why a small local model instead of a bigger cloud one?

Free tools only, and customer numbers stay on my own machine. The trade off is that a small model makes more slips, which is why I built the checks.

Recommendations and the A/B test

Show you can move from analysis to action.

What are your three recommendations?

One: contact big spenders who are slipping first, using the risk times spend list. Two: get new customers to buy a second time, since only 42.3% come back in 180 days. Three: push the fashion newsletter sign up.

I sized each as a what if. Middle cases protect about 1.2%, 0.9% and 0.05% of yearly revenue, with ranges of 0.6% to 1.8%, 0.4% to 1.9% and 0.03% to 0.11%.

How did you size them? Are those numbers reliable?

They are what ifs with stated assumptions, not forecasts. For idea one I assumed the offer saves 10%, 20% or 30% of the revenue the list catches. For idea two I assumed 2, 5 or 10 more points of new customers come back. For idea three I assumed 10% or 20% of non subscribers sign up and only a quarter or half of the gap is real.

Ideas one and two overlap, so I say that out loud. Idea one ranks first because its starting pool is measured. The newsletter ranks last even though it looked best in the raw percentages, because most of that gap is people who already like H&M.

Design an A/B test for recommendation one.

Question: does one personalised email with a modest offer raise revenue per customer over the next 90 days, compared with doing nothing?

Who: established customers in the top 20% of the risk times spend list. Split: randomly, 50 and 50, from that same list. Treatment gets the email and offer, control gets nothing, and no other campaign differs between them. Before launch I check both groups look alike on age, past spend and days since last purchase.

Main result: revenue per customer in the next 90 days, after the cost of the discount. Also watched: the share who bought at least once, and unsubscribes as a safety check.

Why a random split, and not the people who opened old emails?

People who open emails already like the brand and would probably buy anyway. If they buy after my offer I cannot tell if the offer worked. Random assignment makes both groups equally keen and equally at risk, so any gap afterwards comes from the offer. That is what turns linked to into caused by.

How many people and how long?

About 4,300 per group to detect a 3 point rise in the share who buy, assuming around 58% buy in the control group. That is a rough estimate. Revenue per customer is lopsided because a few people spend a lot, so I would size the final test with the real spread from past data. Run it for 90 days after sending plus a short buffer, and no peeking to stop early when it looks good.

What would make you stop, and what if the result is flat?

Stop if unsubscribes jump, or if profit after the discount is negative. Roll out only if revenue per customer is higher and the 95% margin stays above zero after the discount cost. If the margin includes zero, it is not proven, so I would not roll out, and I would ask whether the test was too small or the offer too weak.

Tough questions

The ones that decide the interview. Answer them honestly and calmly.

What is the weakest part of this project?

Several things, and I would rather tell you than have you find them. I used a 10% sample. The prices are scrambled, so there are no pounds. The data ends in 2020, partly in the pandemic. There are no costs or margins. My definition of churn, 90 days without a purchase, is a proxy and not proof someone left. And the whole analysis shows links, not causes. That last point is exactly why I designed the A/B test.

What would you do with more time?

Run everything on the full data. Bring in the product table, which I did not use, to see which categories hold customers. Build a proper customer lifetime value model, since I used past year spend as a simple stand in. And run the A/B test for real.

Did you do customer lifetime value?

Not a full model. I used each customer's past 365 day spend as a simple measure of value, and I say so. A proper lifetime value model is the natural next step.

How did you use AI tools? Did you write this yourself?

I used Claude as a coach and pair programmer. The scripts were drafted with its help. I ran every step myself in PyCharm, challenged the results, made the decisions, like the loyal cut off and the risk times spend ranking, and wrote my reasoning in a decision log. I can walk you through any line and tell you why it is there.

> Be honest, and only say this if it is true for you. Then be ready to explain any step, because they may ask you to open the code.

Explain the project to a manager in three sentences.

A fifth of yearly revenue sits with customers who have gone quiet or are drifting, and about 12% is probably gone for good. The model is good at finding who is at risk, but ranking by risk times spend finds the people worth the most money, about half the revenue that is lost from the top 20% of the list. Before spending on it, we should run a randomised test, because everything I found so far shows links, not proof.

Write a query to find each customer's second shopping day.

I would say it out loud first: one row per customer and day, rank the days, keep rank 2.

```
SELECT customer_id, min(t_dat) AS second_day
FROM (
  SELECT customer_id, t_dat,
         dense_rank() OVER (PARTITION BY customer_id ORDER BY t_dat) AS rk
  FROM (SELECT DISTINCT customer_id, t_dat FROM sample_transactions)
)
WHERE rk = 2
GROUP BY customer_id;
```

The inner SELECT DISTINCT gives one row per customer and day. dense_rank numbers each customer's days from 1. rk = 2 keeps the second visit. dense_rank, not row_number, so two items on the same day never count as two visits.

Things never to say

Each of these sounds fine and gets picked apart in one follow up. Say the right version instead.

The model is 77% accurate.	AUC is not accuracy. Say AUC 0.77, and explain it as the leaver ranked above the stayer.
The newsletter increases repeat purchases.	Say linked to. Selection explains much of it, and only a random test shows cause.
H&M is losing 21.5% of revenue.	Say up to 21.5% is exposed, about 12.1% is likely gone for good.
Customers left because of Zara and Shein.	The data shows who went quiet, not why. Say that.
Revenue is in pounds.	It is an index. Prices in the data are scrambled.
I built it in Power BI.	You used Tableau. Say so and say why.
The contact list saves 49.4% of lost revenue.	It catches 49.4% in a test on past data. Whether contact saves it is for the A/B test.

Questions to ask them

How does your team decide who gets a retention offer today, and how do you know it works?

Do you run randomised tests on campaigns, or mostly compare before and after?

Which part of the customer journey do you think leaks most, first purchase to second, or later?

How do analysts here share findings with marketing, as dashboards, written notes or meetings?

What would a strong first three months look like in this role?

Phrases to keep ready

Short lines that make you sound careful and sure at the same time.

Linked to, not caused by.

I used it to rank customers, not to quote exact odds.

Exposed, up to 21.5%. Likely lost, about 12.1%.

Here is what I checked before I trusted it.

I do not know, here is my guess, and here is how I would find out.

A better predictor can protect less money.

I would test it with a random split before spending on it.

Data: H&M Personalized Fashion Recommendations (Kaggle), used for non-commercial learning. No raw data is shared. Prices in the data are scrambled, so revenue is an index.